



NCI Protocol #: 10323

Patient ID:

Date:

1 of 13

Patient Name:

Date Collected:

Biopsy Site: **Bone Marrow Aspirate**

Sample ID:

Telephone:

Tumor Content (%): **95%**

Patient DOB:

Referring Physician:

Primary Diagnosis: **Acute Myeloid Leukemia**

MoCha ID:

Fax:

Preanalytics to include tumor enrichment as required is performed by Van Andel Research Institute; Grand Rapids; MI 49503

Relevant Acute Myeloid Leukemia Findings

Gene	Finding	Gene	Finding
ABL1	None detected	MECOM	None detected
ASXL1	None detected	MLLT3	None detected
CEBPA	None detected	MYH11	None detected
CREBBP	None detected	NPM1	NPM1 W288Cfs*12
FLT3	FLT3 ITD mutation	NUP214	None detected
IDH1	None detected	RARA	None detected
IDH2	None detected	RUNX1	None detected
KMT2A	None detected	TP53	None detected

Relevant Biomarkers

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	FLT3 ITD mutation	gilteritinib ¹ midostaurin + chemotherapy ¹ cytarabine + daunorubicin cytarabine + daunorubicin + etoposide cytarabine + etoposide + idarubicin cytarabine + fludarabine + idarubicin + filgrastim cytarabine + idarubicin cytarabine + mitoxantrone gemtuzumab ozogamicin + chemotherapy sorafenib sorafenib + chemotherapy	None	38
Prognostic significance: ELN 2017: Intermediate				

Public data sources included in relevant therapies: FDA¹, NCCN

Public data sources included in prognostic and diagnostic significance: NCCN

Tier Reference: Li et al. *Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists.* J Mol Diagn. 2017 Jan;19(1):4-23.

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.10(005). The content of this report has not been evaluated or approved by FDA, EMA or other regulatory agencies.

Relevant Biomarkers (continued)

Tier	Genomic Alteration	Relevant Therapies (In this cancer type)	Relevant Therapies (In other cancer type)	Clinical Trials
IA	NPM1 W288Cfs*12	cytarabine + daunorubicin cytarabine + daunorubicin + etoposide cytarabine + etoposide + idarubicin cytarabine + fludarabine + idarubicin + filgrastim cytarabine + idarubicin cytarabine + mitoxantrone gemtuzumab ozogamicin + chemotherapy midostaurin + chemotherapy	None	1
Prognostic significance: ELN 2017: Intermediate				

Public data sources included in relevant therapies: FDA1, NCCN
Public data sources included in prognostic and diagnostic significance: NCCN
Tier Reference: Li et al. Standards and Guidelines for the Interpretation and Reporting of Sequence Variants in Cancer: A Joint Consensus Recommendation of the Association for Molecular Pathology, American Society of Clinical Oncology, and College of American Pathologists. J Mol Diagn. 2017 Jan;19(1):4-23.

Prevalent cancer biomarkers without relevant evidence based on included data sources

PTPN11 F71L, PTPN11 G503A, TET2 W1847*, TET2 R544*

Variant Details

DNA Sequence Variants								
Gene	Amino Acid Change	Coding	Variant ID	Locus	Allele Frequency	Coverage	Transcript	Variant Effect
PTPN11	p.(F71L)	c.211T>C	COSM13039	chr12:112888195	5.21%	1998	NM_002834.5	missense
PTPN11	p.(G503A)	c.1508G>C	COSM13027	chr12:112926888	15.20%	2000	NM_002834.5	missense
FLT3	p.(F594_D600dup)	c.1775_1775delTinsT . TGATTTTCAGAGAAT ATGAATA		chr13:28608281	24.60%		NM_004119.3	nonframeshift Block Substitution
TET2	p.(R544*)	c.1630C>T	COSM41850	chr4:106156729	48.22%	1995	NM_001127208.2	nonsense
TET2	p.(W1847*)	c.5541G>A	.	chr4:106197208	45.07%	1999	NM_001127208.2	nonsense
NPM1	p.(W288Cfs*12)	c.863_864insTCTG	COSM17559	chr5:170837547	33.54%	1980	NM_002520.6	frameshift Insertion

Low Coverage Regions:

N/A

Comments:

N/A

Report Signed By

Name	Role	Date

Biomarker Descriptions

FLT3 (fms related receptor tyrosine kinase 3)

Background: The FLT3 gene encodes the fms related tyrosine kinase 3, a tyrosine kinase receptor that is a member of the class III receptor tyrosine kinase family that also includes PDGFR, FMS, and KIT¹. FLT3 is highly expressed in hematopoietic progenitor cells². Genomic alterations in FLT3 activate downstream oncogenic pathways including PI3K/AKT/mTOR and RAS/RAF/MEK/ERK pathways which promote cellular proliferation, survival, and inhibition of differentiation¹.

Alterations and prevalence: Somatic mutations occur in approximately 30% of acute myeloid leukemia (AML), 7-10% of melanoma, and up to 8% of uterine cancer^{3,4,5,6}. The most common activating FLT3 mutations are internal tandem duplications (ITD) that range from 3 to 400 base pairs in length within exons 14 and 15 in the juxtamembrane (JM) domain⁷. The second most frequent mutations are point mutations in exon 20 within the tyrosine kinase domain (TKD)⁸. FLT3 is amplified in up to 8% of colorectal cancer, 3% of stomach cancer, and is commonly overexpressed in AML^{5,6,9}.

Potential relevance: The presence of FLT3-ITD confers poor prognosis in myelodysplastic syndrome (MDS) and AML^{10,11}. Similarly, the FLT3 TKD mutation D835 confers poor prognosis in MDS¹⁰. Midostaurin¹² (2017) and gilteritinib¹³ (2018) are kinase inhibitors approved for AML patients with FLT3-ITD and TKD mutations including D835 and I836 mutations. The FDA granted fast track designations in 2017 to crenolanib¹⁴ for FLT3 mutation-positive relapsed or refractory AML and in 2018 to quizartinib¹⁵ for AML with FLT3-ITD. A phase II trial testing crenolanib in 34 patients with FLT3-ITD and TKD mutated relapsed/refractory AML, reported that FLT3 inhibitor naïve patients demonstrated a longer overall survival (OS) and event free survival (EFS) in comparison to previously treated patients (median OS: 55 weeks vs 13 weeks; median EFS: 13 weeks vs 7 weeks)¹⁶. Another phase II trial of crenolanib with chemotherapy in newly diagnosed FLT3 mutated AML reported complete remission in 24/29 (83%) patients¹⁷. Several multi-targeted tyrosine kinase inhibitors such as sorafenib (2005), sunitinib (2006), cabozantinib (2012), and ponatinib (2012) are FDA approved and include FLT3 as a target. Sorafenib is recommended in combination with chemotherapy in FLT3-ITD mutated AML¹¹.

NPM1 (nucleophosmin 1)

Background: The NPM1 gene encodes the nucleophosmin protein, a histone chaperone of the nucleophosmin/nucleoplasmin family, which also includes NPM2 and NPM3¹⁸. NPM1 functions as an oncogene and tumor suppressor, and is important in maintaining genomic stability, DNA repair, and apoptosis^{18,19}. NPM1 has a highly conserved N-terminal region which constitutes the core domain responsible for oligomerization, an acidic domain, a nuclear localization signal, and a disorganized C-terminal region which is required for nucleolar localization¹⁸. Oligomerization of NPM1 localizes the protein in the nucleus of proliferating cells where it binds to Akt in response to growth factor stimulation and escapes proteolytic degradation by caspase activity, thereby promoting cell survival^{18,19}. NPM1 is one of the most frequently altered genes in hematological cancers²⁰. Most NPM1 mutations occur in the C-terminus, impacting protein folding or the nucleolar localization signal, and result in the localization of NPM1 to the cytoplasm (NPMc) instead of to the nucleus¹⁸.

Alterations and prevalence: NPM1 mutations are observed in 45-60% of AML with a normal karyotype (NK-AML), 28-35% of de novo acute myeloid leukemia (AML) and are frequently co-mutated with DNMT3A and/or FLT3-ITD^{11,21,22}. NPM1 fusions are associated with distinct partner genes in acute promyelocytic leukemia (APL), anaplastic large-cell lymphoma (ALCL), AML, and myelodysplasia²⁰. Specifically, NPM1-ALK fusion is found in 30% of all ALCL and this specific fusion is observed in 85% of ALK-positive ALCL¹⁸. The t(5;17)(q35;q21) translocation that results in NPM1-RARA fusion is observed in APL²³.

Potential relevance: NPM1 mutated AML is recognized as a distinct diagnostic disease entity by the World Health Organization (WHO)²⁴. NPM1 mutations are associated with better outcomes, increased complete remission, and improved overall survival in AML^{11,22}. NPM1 without FLT3-ITD mutations or with <0.5 allelic ratio FLT3-ITD mutations are associated with favorable risk in AML¹¹. Concurrent NPM1 and with >0.5 allelic ratio FLT3-ITD mutations confer intermediate risk in AML, whereas wild-type NPM1 confers poor/adverse risk¹¹. The NPM1 frameshift mutation W288fs*12 is associated with poor prognosis in myelodysplastic syndrome

Biomarker Descriptions (continued)

(MDS)¹⁰. The ALK-NPM1 fusion, and translocation t(2;5)(p23;q35) which leads to an ALK-NPM1 fusion, is diagnostic of ALK-positive anaplastic large cell lymphoma^{25,26}.

PTPN11 (protein tyrosine phosphatase non-receptor type 11)

Background: The PTPN11 gene encodes a tyrosine phosphatase non-receptor type 11 protein, and is also known as Src homology region 2 domain-containing phosphatase-2 (SHP-2)²⁷. PTPN11 is a member of the protein tyrosine phosphatase (PTP) family that is ubiquitously expressed and regulates cellular growth, differentiation, mitotic cycle, and oncogenic transformation. PTPN11 contains two tandem N-terminal Src homology-2 domains (N-SH2 and C-SH2), a PTP catalytic domain, and uncharacterized C-terminal domain²⁸. PTPN11 regulates various signaling processes including the RAS/RAF/MEK/ERK, PI3K/AKT/MTOR, and JAK/STAT pathways^{29,30}. Germline mutations in PTPN11 are associated with LEOPARD syndrome and Noonan syndrome with a predisposition to juvenile myelomonocytic leukemia (JMML) or myeloproliferative neoplasms (MPN)^{10,31}. Somatic mutations in PTPN11 are associated with JMML^{32,33} and solid tumors such as lung, colon, and thyroid^{28,34}.

Alterations and prevalence: Somatic alterations in PTPN11 include mutations and amplification^{31,35}. PTPN11 mutations occur in 6% of uterine carcinoma and 5% of acute myeloid leukemia (AML) cases⁵. Mutations including E76K and D61Y result in PTPN11 activation and are associated with 30% of JMML³⁰.

Potential relevance: Currently, no therapies are approved for PTPN11 aberrations. Somatic mutations in PTPN11 confer drug resistance to venetoclax and azacitidine in AML^{36,37}.

TET2 (tet methylcytosine dioxygenase 2)

Background: TET2 encodes the tet methylcytosine dioxygenase 2 protein and belongs to a family of ten-eleven translocation (TET) proteins that also includes TET1 and TET3³⁸. TET2 is involved in DNA methylation, specifically in the conversion of 5-methylcytosine to 5-hydroxymethylcytosine^{39,40}. The TET proteins contain a C-terminal core catalytic domain that contains a cysteine-rich domain and a double stranded β -helix domain (DSBH)⁴¹. TET2 is a tumor suppressor gene. Loss of function mutations in TET2 are associated with loss of catalytic activity and transformation to hematological malignancies^{38,39,40}.

Alterations and prevalence: Somatic TET2 mutations, including nonsense, frameshift, splice site, and missense, are observed in 20-25% of myelodysplastic syndrome (MDS) associated diseases, including 40%-60% chronic myelomonocytic leukemia (CMML)¹⁰. TET2 mutations at H1881 and R1896 are frequently observed in myeloid malignancies^{39,42}. TET2 mutations are also observed in 9% of uterine, 8% of melanoma and acute myeloid leukemia (AML), as well as 6% of diffuse large B-cell lymphoma (DLBCL).

Potential relevance: The presence of TET2 mutations may be used as one of the major diagnostic criteria in pre-primary myelofibrosis (pre-PMF) and overt PMF in the absence of JAK2/CALR/MPL mutations^{24,43}. TET2 mutations are associated with poor prognosis in PMF and increased rate of transformation to leukemia^{43,44}.

Relevant Therapy Summary

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

FLT3 ITD mutation

Relevant Therapy	FDA	NCCN	Clinical Trials*
gilteritinib	☒	☒	☒
midostaurin + cytarabine + daunorubicin	☒	☒	☒

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.10(005).

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

FLT3 ITD mutation (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
cytarabine + daunorubicin	×	●	×
cytarabine + daunorubicin + etoposide	×	●	×
cytarabine + etoposide + idarubicin	×	●	×
cytarabine + fludarabine + idarubicin + filgrastim	×	●	×
cytarabine + idarubicin	×	●	×
cytarabine + mitoxantrone	×	●	×
gemtuzumab ozogamicin + cytarabine + daunorubicin	×	●	×
midostaurin + cytarabine	×	●	×
sorafenib	×	●	×
sorafenib + azacitidine	×	●	×
sorafenib + decitabine	×	●	×
chemotherapy, midostaurin	×	×	● (III)
crenolanib, chemotherapy	×	×	● (III)
crenolanib, midostaurin, chemotherapy	×	×	● (III)
gemtuzumab ozogamicin, chemotherapy, gilteritinib	×	×	● (III)
gilteritinib, midostaurin, chemotherapy	×	×	● (III)
SKLB1028	×	×	● (III)
chemotherapy	×	×	● (II/III)
chemotherapy, venetoclax	×	×	● (II/III)
ibrutinib, sorafenib, chemotherapy	×	×	● (II/III)
alemtuzumab, chemotherapy	×	×	● (II)
allogeneic stem cells, chemotherapy	×	×	● (II)
chemotherapy, gilteritinib, midostaurin	×	×	● (II)
chemotherapy, radiation therapy	×	×	● (II)
HYML-122	×	×	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

FLT3 ITD mutation (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
HYML-122, chemotherapy	×	×	● (II)
ponatinib	×	×	● (II)
radiation therapy, allogeneic stem cells, chemotherapy	×	×	● (II)
venetoclax, chemotherapy	×	×	● (II)
CC-90009, gilteritinib	×	×	● (I/II)
dubermatinib, chemotherapy	×	×	● (I/II)
gilteritinib, chemotherapy	×	×	● (I/II)
navtemadlin, TL-895	×	×	● (I/II)
NMS-P948	×	×	● (I/II)
ponatinib, chemotherapy	×	×	● (I/II)
quizartinib, chemotherapy	×	×	● (I/II)
stem cell therapy	×	×	● (I/II)
venetoclax, gilteritinib, chemotherapy	×	×	● (I/II)
allogeneic double negative T cells	×	×	● (I)
clifutinib	×	×	● (I)
gemtuzumab ozogamicin, midostaurin, chemotherapy	×	×	● (I)
LNK01002	×	×	● (I)
nintedanib, chemotherapy	×	×	● (I)
PHI-101	×	×	● (I)
siremadlin, midostaurin	×	×	● (I)

NPM1 W288Cfs*12

Relevant Therapy	FDA	NCCN	Clinical Trials*
cytarabine + daunorubicin	×	●	×
cytarabine + daunorubicin + etoposide	×	●	×

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Relevant Therapy Summary (continued)

☒ In this cancer type
 ☐ In other cancer type
 ☒ In this cancer type and other cancer types
 ☒ No evidence

NPM1 W288Cfs*12 (continued)

Relevant Therapy	FDA	NCCN	Clinical Trials*
cytarabine + etoposide + idarubicin	×	●	×
cytarabine + fludarabine + idarubicin + filgrastim	×	●	×
cytarabine + idarubicin	×	●	×
cytarabine + mitoxantrone	×	●	×
gemtuzumab ozogamicin + cytarabine + daunorubicin	×	●	×
midostaurin + cytarabine + daunorubicin	×	●	×
venetoclax, chemotherapy	×	×	● (II)

* Most advanced phase (IV, III, II/III, II, I/II, I) is shown and multiple clinical trials may be available.

Clinical Trials Summary

FLT3 ITD mutation + NPM1 W288Cfs*12

NCT ID	Title	Phase
NCT04867928	A multicentric phase II study of venetoclax and azacitidine for the management of the molecular relapse/progression in adult NPM1- mutated acute myeloid leukemia.	II

FLT3 ITD mutation

NCT ID	Title	Phase
NCT04174612	A Phase III, Prospective, Randomized Multi-center Intervention Trial of Early Intensification in AML Patients Bearing FLT3 Mutations Based on Peripheral Blast Clearance: A MYNERVA-GIMEMA Study	III
NCT03250338	Phase III Randomized, Double-blind, Placebo-controlled Study Investigating the Efficacy of the Addition of Crenolanib to Salvage Chemotherapy Versus Salvage Chemotherapy Alone in Subjects < or = 75 Years of Age With Relapsed/Refractory FLT3 Mutated Acute Myeloid Leukemia	III
NCT03258931	Phase III Randomized Study of Crenolanib Versus Midostaurin Administered Following Induction Chemotherapy and Consolidation Therapy in Newly Diagnosed Subjects With FLT3 Mutated Acute Myeloid Leukemia	III
NCT04293562	A Phase III Randomized Trial for Patients With De Novo AML Comparing Standard Therapy Including Gemtuzumab Ozogamicin (GO) to CPX-351 With GO, and the Addition of the FLT3 Inhibitor Gilteritinib for Patients With FLT3 Mutations	III

Clinical Trials Summary (continued)

FLT3 ITD mutation (continued)

NCT ID	Title	Phase
NCT04027309	A Phase III, Multicenter, Open-label, Randomized, Study of Gilteritinib Versus Midostaurin in Combination With Induction and Consolidation Therapy Followed by One-year Maintenance in Patients With Newly Diagnosed Acute Myeloid Leukemia (AML) or Myelodysplastic Syndromes With Excess Blasts-2 (MDS-EB2) With FLT3 Mutations Eligible for Intensive Chemotherapy (HOVON 156 AML / AMLSG 28-18)	III
NCT04716114	A Phase III, Open-Label, Multicenter, Randomized Study of SKLB1028 Versus Salvage Chemotherapy in Patients With FLT3-mutated Acute Myeloid Leukemia Refractory to or Relapsed After First-line Treatment(ALIVE)	III
NCT03256071	Low Dose Decitabine + Modified BUCY Conditioning Regimen for High Risk Acute Myeloid Leukemia Undergoing Allo-HSCT	II/III
NCT02416388	Phase II/III Randomized Study to Improve Overall Survival in 18 to 60 Year-old Patients, Comparing Daunorubicin Versus High Dose Idarubicin Induction Regimens, High Dose Versus Intermediate Dose Cytarabine Consolidation Regimens, and Standard Versus Mycophenolate Mofetil Prophylaxis of Graft Versus Host Disease in Allografted Patients in First CR : a Backbone InterGroup-1 Trial	II/III
NCT03642236	Combination of Brutons Tyrosine Kinase (BTK) Inhibitor Overcomes Drug-resistance in Refractory/ Relapsed FLT3 Mutant Acute Myeloid Leukemia (AML)	II/III
NCT02626715	A Phase II Study of Myeloablative and Reduced-Intensity Conditioning Regimens for Children and Young Adults With Acute Myeloid Leukemia or Myelodysplastic Syndrome Undergoing Allogeneic Hematopoietic Stem Cell Transplantation.	II
NCT01760655	A Two Step Approach to Reduced Intensity Allogeneic Hematopoietic Stem Cell Transplantation for High Risk Hematologic Malignancies	II
NCT02115295	Phase II Study of Cladribine Plus Idarubicin Plus Cytarabine (ARAC) in Patients With AML, HR MDS, or Myeloid Blast Phase of CML	II
NCT03121014	A Phase II Study of Intensity Modulated Total Marrow Irradiation (IM-TMI) in Addition to Fludarabine/ Busulfan Conditioning for Allogeneic Transplantation in High Risk AML and Myelodysplastic Syndromes	II
NCT03836209	Randomized Trial of Gilteritinib vs Midostaurin in FLT3 Mutated Acute Myeloid Leukemia (AML).	II
NCT05241106	A Single-arm, Open, Multicenter, Phase II Study to Investigator the Efficacy and Safety of HYML-122 in Patients With FLT3 Positive Relapsed or Refractory Acute Myeloid Leukemia (AML)	II
NCT05241093	A Single-arm, Open, Multicenter, Phase II Study to Investigator the Efficacy and Safety of HYML-122 and Cytarabine in Patients With FLT3 Positive Relapsed or Refractory Acute Myeloid Leukemia (AML)	II
NCT03690115	Phase II Study of Ponatinib (Iclusig) for Prevention of Relapse After Allogeneic Stem Cell Transplantation (Allo-SCT) in FLT3-ITD AML Patients: the PONALLO Trial."	II
No NCT ID	Phase II Study Of Allogeneic Hematopoietic Cell Transplantation For Children With Acute Myeloid Leukemia In First And Second Complete Remission Using Fludarabine, Cytarabine, Melphalan And Low-Dose Total Body Irradiation As A Conditioning Regimen (AML-SCT15).	II
NCT04336982	An Exploratory Phase I/II Open-Label Multi-Arm Trial to Evaluate the Safety and Efficacy of CC-90009 Combinations in Subjects With Acute Myeloid Leukemia	I/II
NCT04518345	A Phase Ib/II Study of TP-0903 in Patients With Acute Myeloid Leukemia and FLT3 Mutations	I/II

Clinical Trials Summary (continued)

FLT3 ITD mutation (continued)

NCT ID	Title	Phase
NCT04240002	A Phase I/II, Multicenter, Open-Label, Single Arm, Dose Escalation and Expansion Study of Gilteritinib (ASP2215) Combined With Chemotherapy in Children, Adolescents and Young Adults With FMS-like Tyrosine Kinase 3 (FLT3)/Internal Tandem Duplication (ITD) Positive Relapsed or Refractory Acute Myeloid Leukemia (AML)	I/II
NCT04669067	An Open-Label, Multicenter, Phase Ib/II Study of the Safety and Efficacy of TL-895 Combined With KRT-232 in Patients With Relapsed/Refractory (R/R) FLT3+ Acute Myeloid Leukemia (AML)	I/II
NCT03922100	A Phase I/II Study of NMS-03592088, a FLT3, KIT and CSF1R Inhibitor, in Patients With Relapsed or Refractory AML or CMML	I/II
NCT02829840	Phase I/II, Dose-Escalation Study of Ponatinib, a FLT3 Inhibitor, With and Without Combination of 5-Azacitidine, in Patients With FLT3-Mutated Acute Myeloid Leukemia (AML)	I/II
NCT03793478	A Phase I/II, Multicenter, Dose-Escalating Study To Evaluate the Safety, Pharmacokinetics, Pharmacodynamics, and Efficacy Of Quizartinib Administered in Combination with Re-Induction Chemotherapy, and as a Single-Agent Continuation Therapy, in Pediatric Relapsed/Refractory AML Subjects Aged 1 Month to <18 Years (and Young Adults Aged up to 21 Years) with FLT3-ITD mutations.	I/II
NCT01203722	Reduced Intensity, Partially HLA Mismatched Allogeneic BMT for Hematologic Malignancies Using Donors Other Than First-degree Relatives	I/II
NCT04140487	A Phase I/II Study of Azacitidine, Venetoclax, and Gilteritinib for Patients With Acute Myeloid Leukemia or High-Risk Myelodysplastic Syndrome With an Activating FLT3 Mutation	I/II
NCT05010122	A Phase I/II Study of ASTX727, Venetoclax, and Gilteritinib for Patients With Acute Myeloid Leukemia or High-Risk Myelodysplastic Syndrome With an Activating FLT3 Mutation	I/II
No NCT ID	Safety and Effective of Prophylactic Infusion of DNT cells in Allogeneic Hematopoietic Stem Cell Transplantation for Patients with Myeloid Malignancies: A Single Center Clinical Research	I
No NCT ID	A Phase I Study to Assess the Safety of Micro-dose Lenalidomide as Maintenance Therapy Post-allogeneic Haematopoietic Cell Transplantation for Patients with Acute Myeloid Leukaemia or Myelodysplastic Syndromes, at High Risk of Relapse	I
NCT04827069	A Phase I, Multi-center, Open,Single Arm, Dose-escalation Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Efficacy of Clifutinib Besylate(HEC73543) in Relapsed or Refractory Acute Myeloid Leukemia (AML)	I
NCT03900949	A Phase I Study to Evaluate the Safety and Tolerability of Gemtuzumab Ozogamicin and Midostaurin When Used in Combination With Standard Cytarabine and Daunorubicin Induction for Newly Diagnosed FLT3-mutated Acute Myeloid Leukemia	I
NCT05024552	A Phase I Study of Vyxeos Plus Gilteritinib in Relapsed or Refractory, FLT3-Mutated Acute Myeloid Leukemia	I
NCT04896112	An Open-Label, Multicenter, Phase I Study to Evaluate the Safety, Pharmacokinetics and Preliminary Efficacy of LNK01002 in Patients With Malignant Myeloid Hematologic Neoplasms	I
NCT03513484	Phase Ib, Open Label, Combination Study of Nintedanib With 5-Azacitidine in Acute Myeloid Leukemia Characterized by HOX Gene Overexpression, That Are Not Candidates of Intensive Chemotherapy	I

Clinical Trials Summary (continued)

FLT3 ITD mutation (continued)

NCT ID	Title	Phase
NCT04842370	A Prospective, Phase Ia/Ib, Multicenter Study to Evaluate the Safety, Tolerability, Pharmacokinetics and Pharmacodynamics of the FLT3 Inhibitor, PHI 101, Alone in Subjects With Relapsed or Refractory Acute Myeloid Leukemia (AML).	I
NCT04496999	HDM201 and Midostaurin (HDMM) in Relapsed/Refractory AML With FLT3mut and TP53wt; a Phase I Study.	I

Assay Information

Methodology, Scope, and Application: The NCI Myeloid Assay v2 (NMAv2) is a next-generation sequencing (NGS) assay which identifies pre-defined and novel genomic variants covering 45 DNA genes and 35 fusion drivers that are categorized into 3 mutation types: single nucleotide variants (SNVs), insertions/deletions (Indels) including FLT3 internal tandem duplications (ITDs), and gene fusions. The assay utilizes Thermo Fisher Scientific's Oncomine® Myeloid Assay GX v2, a NGS assay that utilizes a multiplex polymerase chain reaction (PCR) with DNA and RNA extracted from blood and bone marrow mononuclear cells for sequencing on the Ion Torrent Genexus Integrated Sequencer and analyzed by the Ion Torrent Genexus pipeline version 6.6.2.1. The NMAv2 can reliably identify the presence or absence of 1661 hotspot variants and 779 reported fusions compared to the Human Reference Genome hg19, including the genes listed below. The NMAv2 is a laboratory developed test designed to detect gene mutations for major myeloid disorders.

Analytical Sensitivity and Specificity: The NMAv2 has been determined to be suitably analytically sensitive and specific for the various types of abnormalities within its reportable range, demonstrating 98.97% sensitivity, 100% specificity and an overall reproducibility with a mean positive percent agreement of 99% and a mean negative percent agreement of 100% during validation of the assay.

Hereditary and Germline Mutations: This assay examines cancer cells only and does not examine normal (non-cancer) cells. Mutations detected by the assay may be present only in the cancer cells, or in every cell of the body (including non-cancer cells). This test also cannot tell whether a potential germline mutation causes or will cause a hereditary cancer syndrome. If the patient's history includes any one or more features suggestive of a possible hereditary cancer predisposition (such as, cancer arising at a young age, especially a cancer of a type unusual in younger patients; a personal history of multiple different types of cancers; or a significant cancer history among blood relatives, especially but not exclusively of the same type as the patient's), it is recommended that the physician arrange for the patient to meet with a genetic counselor and, if warranted, undergo the appropriate genetic test on normal (i.e. non-cancer) tissue (blood or cells brushed from the oral surface of the cheek) to check for a germline abnormality, regardless of the results of this research study.

Report Content: Thermo Fisher Scientific's Ion Torrent Oncomine Knowledgebase Reporter (OKR) software was used in the generation of this report. Software was developed and designed internally by Thermo Fisher Scientific. The analysis was based on the current version of OKR. The data presented here is from a curated knowledgebase of publicly available information but may not be exhaustive. FDA information was sourced from www.fda.gov and is current as of this version. NCCN information was sourced from www.nccn.org and reflects its current version. Clinical Trials information reflects the current version. For the most up-to-date information regarding particular trials, search www.clinicaltrials.gov by NCT ID or search local clinical trials authority website by local identifier listed in 'Other Identifiers.' Variants are reported according to HGVS nomenclature and classified following AMP/ASCO/CAP guidelines (Li et al. 2017). Based on the data sources selected, variants, therapies, and trials identified in this report are listed in order of potential clinical significance but not for predicted efficacy of the therapies.

Genes assayed for hotspot mutations:

ABL1, ANKRD26, BRAF, CBL, CSF3R, DDX41, DNMT3A, FLT3, GATA2, HRAS, IDH1, IDH2, JAK2, KIT, KRAS, MPL, MYD88, NPM1, NRAS, PPM1D, PTPN11, SETBP1, SF3B1, SMC1A, SMC3, SRSF2, U2AF1, WT1

Genes assayed with full exon coverage:

ASXL1, BCOR, CALR, CEBPA, ETV6, EZH2, IKZF1, NF1, PHF6, PRPF8, RB1, RUNX1, SH2B3, STAG2, TET2, TP53, ZRSR2

Shahanawaz Jiwani, MD, PhD, Laboratory Director | CLIA ID 21D2097127

Disclaimer: The data presented here is from a curated knowledgebase of publicly available information, but may not be exhaustive. The data version is 2022.10(005).

Genes assayed for detection of fusions:

ABL1, ABL2, BCL2, BRAF, CCND1, CREBBP, EGFR, ETV6, FGFR1, FGFR2, FUS, HMGA2, JAK2, KAT6A, KAT6B, KMT2A, KMT2A-PTD (Partial Tandem Duplication), MECOM, MET, MLLT10, MRTFA (MKL1), MYBL1, MYH11, NTRK2, NTRK3, NUP214, NUP98, PAX5, PDGFRA, PDGFRB, RARA, RUNX1, TCF3, TFE3, ZNF384

Limitations of the assay

1. Mutations in homopolymer regions of >5 nucleotides may not be accurately assessed by this assay.
2. FLT3-ITDs greater than 120bps may not be detected by this assay. Sensitivity for larger FLT3-ITD's may be reduced and variant allele frequencies may be reported lower than expected.
3. Following genetic alterations will not be reported by this assay: RUNX1 p.Ser445AlafsTer149, ASXL1 p.G646Wfs*12, PRPF8 p.Arg156GlufsTer28

DISCLAIMER

This assay reports SNV, indel and FLT3-ITD mutations $\geq$ 5% Variant Allele Frequency (VAF). This assay is considered a Laboratory Developed Test (LDT). Its performance characteristics have been determined through extensive testing although it has not been cleared by the US Food and Drug Administration and such approval is not required for clinical implementation. Furthermore, any comments included in the report are strictly interpretive and the opinion of the reviewer. This laboratory is certified under the Clinical Laboratory Improvements Amendments (CLIA) for the performance of high-complexity testing for clinical purposes. The correlative data presented is a result of the curation of published data sources but may not be exhaustive. The content of this report has not been evaluated or approved by the FDA or other regulatory agencies.

References

1. Grafone et al. An overview on the role of FLT3-tyrosine kinase receptor in acute myeloid leukemia: biology and treatment. *Oncol Rev.* 2012 Mar 5;6(1):e8. PMID: 25992210
2. Short et al. Emerging treatment paradigms with FLT3 inhibitors in acute myeloid leukemia. *Ther Adv Hematol.* 2019; 10: 2040620719827310. PMID: 30800259
3. Small. FLT3 mutations: biology and treatment. *Hematology Am Soc Hematol Educ Program.* 2006:178-84. PMID: 17124058
4. Cancer Genome Atlas Research Network. Genomic and epigenomic landscapes of adult de novo acute myeloid leukemia. *N Engl J Med.* 2013 May 30;368(22):2059-74. doi: 10.1056/NEJMoa1301689. Epub 2013 May 1. PMID: 23634996
5. Cerami et al. The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov.* 2012 May;2(5):401-4. PMID: 22588877
6. Weinstein et al. The Cancer Genome Atlas Pan-Cancer analysis project. *Nat. Genet.* 2013 Oct;45(10):1113-20. PMID: 24071849
7. Nakao et al. Internal tandem duplication of the *flt3* gene found in acute myeloid leukemia. *Leukemia.* 1996 Dec;10(12):1911-8. PMID: 8946930
8. Yamamoto et al. Activating mutation of D835 within the activation loop of FLT3 in human hematologic malignancies. *Blood.* 2001 Apr 15;97(8):2434-9. PMID: 11290608
9. Carow et al. Expression of the hematopoietic growth factor receptor FLT3 (STK-1/Flk2) in human leukemias. *Blood.* 1996 Feb 1;87(3):1089-96. PMID: 8562934
10. NCCN Guidelines® - NCCN-Myelodysplastic Syndromes [Version 3.2022]
11. NCCN Guidelines® - NCCN-Acute Myeloid Leukemia [Version 2.2022]
12. https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/207997s0081bledt.pdf
13. https://www.accessdata.fda.gov/drugsatfda_docs/label/2022/211349s0031bl.pdf
14. <https://www.globenewswire.com/news-release/2017/12/01/1216122/0/en/Arog-Pharmaceuticals-Receives-FDA-Fast-Track-Designation-for-Crenolanib-in-Relapsed-or-Refractory-FLT3-Positive-AML.html>
15. https://www.daiichisankyo.com/media_investors/media_relations/press_releases/detail/006896.html
16. Jasleen et al. Results of a Phase II Study of Crenolanib in Relapsed/Refractory Acute Myeloid Leukemia Patients (Pts) with Activating FLT3 Mutations. *Blood.* 124:389
17. Bartus et al. Crenolanib in combination with standard chemotherapy in newly diagnosed FLT3-mutated AML patients. *ASH 2017.* Abstract #566
18. Box et al. Nucleophosmin: from structure and function to disease development. 2016 Aug 24;17(1):19. PMID: 27553022
19. Koike et al. Recruitment of phosphorylated NPM1 to sites of DNA damage through RNF8-dependent ubiquitin conjugates. *Cancer Res.* 2010 Sep 1;70(17):6746-56. PMID: 20713529
20. Sportoletti. How does the NPM1 mutant induce leukemia?. *Pediatr Rep.* 2011 Jun 22;3 Suppl 2:e6. PMID: 22053282
21. Hourigan et al. Accurate Medicine: Indirect Targeting of NPM1-Mutated AML. 2016 Oct;6(10):1087-1089. PMID: 27698101
22. Jain et al. Mutated NPM1 in patients with acute myeloid leukemia in remission and relapse. *Leuk. Lymphoma.* 2014 Jun;55(6):1337-44. PMID: 24004182
23. Redner et al. The t(5;17) acute promyelocytic leukemia fusion protein NPM-RAR interacts with co-repressor and co-activator proteins and exhibits both positive and negative transcriptional properties. *Blood.* 2000 Apr 15;95(8):2683-90. PMID: 10753851
24. Arber et al. The 2016 revision to the World Health Organization classification of myeloid neoplasms and acute leukemia. *Blood.* 2016 May 19;127(20):2391-405. PMID: 27069254
25. NCCN Guidelines® - NCCN-Primary Cutaneous Lymphomas [Version 2.2022]
26. NCCN Guidelines® - NCCN-T-Cell Lymphomas [Version 2.2022]
27. Tartaglia et al. Somatic PTPN11 mutations in childhood acute myeloid leukaemia. *Br. J. Haematol.* 2005 May;129(3):333-9. PMID: 15842656
28. Chan et al. The tyrosine phosphatase Shp2 (PTPN11) in cancer. *Cancer Metastasis Rev.* 2008 Jun;27(2):179-92. PMID: 18286234

References (continued)

29. Steelman et al. Roles of the Raf/MEK/ERK and PI3K/PTEN/Akt/mTOR pathways in controlling growth and sensitivity to therapy-implications for cancer and aging. *Aging (Albany NY)*. 2011 Mar;3(3):192-222. PMID: 21422497
30. Liu et al. Inhibition of the Gab2/PI3K/mTOR signaling ameliorates myeloid malignancy caused by Ptpn11 (Shp2) gain-of-function mutations. *Leukemia*. 2017 Jun;31(6):1415-1422. PMID: 27840422
31. Tartaglia et al. Somatic mutations in PTPN11 in juvenile myelomonocytic leukemia, myelodysplastic syndromes and acute myeloid leukemia. *Nat. Genet.* 2003 Jun;34(2):148-50. PMID: 12717436
32. Tartaglia et al. Mutations in PTPN11, encoding the protein tyrosine phosphatase SHP-2, cause Noonan syndrome. *Nat. Genet.* 2001 Dec;29(4):465-8. PMID: 11704759
33. Tartaglia et al. Diversity and functional consequences of germline and somatic PTPN11 mutations in human disease. *Am. J. Hum. Genet.* 2006 Feb;78(2):279-90. PMID: 16358218
34. Jongmans et al. Cancer risk in patients with Noonan syndrome carrying a PTPN11 mutation. *Eur. J. Hum. Genet.* 2011 Aug;19(8):870-4. PMID: 21407260
35. Liu et al. Gain-of-function mutations of Ptpn11 (Shp2) cause aberrant mitosis and increase susceptibility to DNA damage-induced malignancies. *Proc. Natl. Acad. Sci. U.S.A.* 2016 Jan 26;113(4):984-9. PMID: 26755576
36. Chyla et al. Genetic Biomarkers Of Sensitivity and Resistance to Venetoclax Monotherapy in Patients With Relapsed Acute Myeloid Leukemia. *Am. J. Hematol.* 2018 May 17. PMID: 29770480
37. Zhang et al. Biomarkers Predicting Venetoclax Sensitivity and Strategies for Venetoclax Combination Treatment. *Blood* 2018 132:175; doi: <https://doi.org/10.1182/blood-2018-175>. 132:175
38. Pan et al. The TET2 interactors and their links to hematological malignancies. *IUBMB Life*. 2015 Jun;67(6):438-45. PMID: 26099018
39. Ko et al. Impaired hydroxylation of 5-methylcytosine in myeloid cancers with mutant TET2. *Nature*. 2010 Dec 9;468(7325):839-43. PMID: 21057493
40. Solary et al. The Ten-Eleven Translocation-2 (TET2) gene in hematopoiesis and hematopoietic diseases. *Leukemia*. 2014 Mar;28(3):485-96. PMID: 24220273
41. An et al. TET family dioxygenases and DNA demethylation in stem cells and cancers. *Exp. Mol. Med.* 2017 Apr 28;49(4):e323. PMID: 28450733
42. Kosmider et al. TET2 mutation is an independent favorable prognostic factor in myelodysplastic syndromes (MDSs). *Blood*. 2009 Oct 8;114(15):3285-91. PMID: 19666869
43. NCCN Guidelines® - NCCN-Myeloproliferative Neoplasms [Version 3.2022]
44. Lundberg et al. Clonal evolution and clinical correlates of somatic mutations in myeloproliferative neoplasms. *Blood*. 2014 Apr 3;123(14):2220-8. PMID: 24478400